

# Close the loop

Congestion estimation matters when it changes placement

# The idea

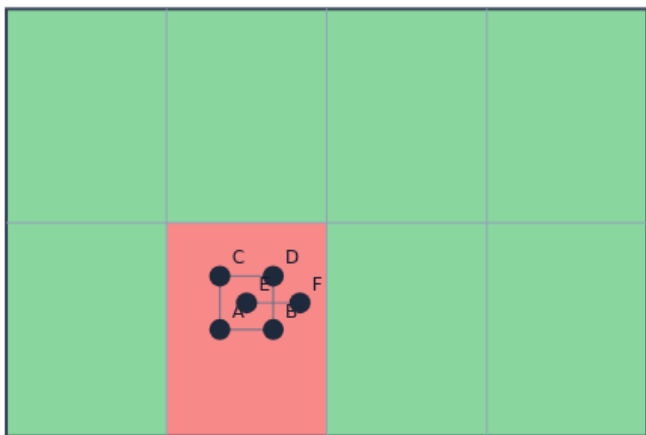
- After inflation
- Clamp to the chip
- Recompute RUDY
- On congested\_seed you should see total overflow drop after one pass

# Hot starter

PLACEMENT FEEDBACK

## Hot starter

Step 1 / 5 · seed · module04-01-placement-feedback



### Explanation

Congested seed starts with total overflow 5 at Cap=2.

- RUDY demand

Need a loop

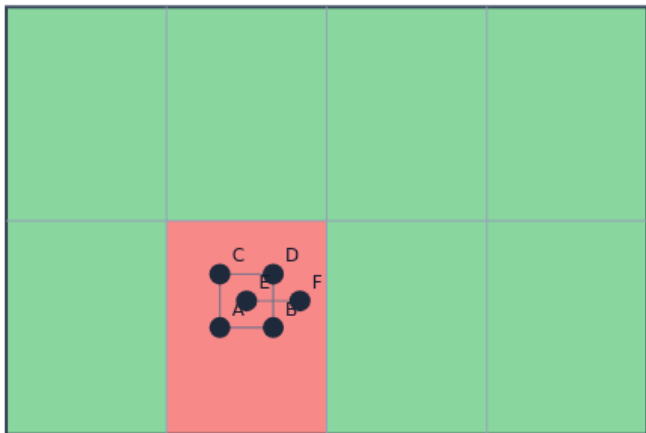
Hot starter

# Estimate

PLACEMENT FEEDBACK

## Estimate

Step 2 / 5 · estimate · module04-01-placement-feedback



### Explanation

Run RUDY → congestion → overflow per tile.

- Same estimators

Matrix in hand

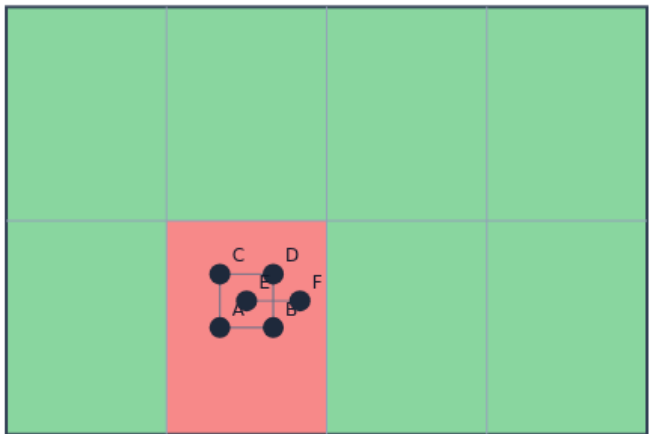
Estimate

# Push outward

PLACEMENT FEEDBACK

## Push outward

Step 3 / 5 · push · module04-01-placement-feedback



### Explanation

Cells in overflowing tiles step toward the quietest neighbor GCell.

- Clamp to chip

Toy one-pass

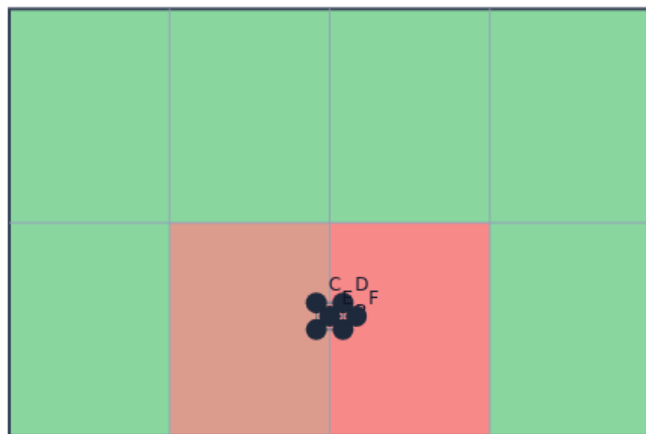
Push outward

# Overflow drops

PLACEMENT FEEDBACK

## Overflow drops

Step 4 / 5 · after · module04-01-placement-feedback



### Explanation

After one feedback pass, total overflow falls (often to ~0 on this toy).

- Remeasure demand

Assert after < before

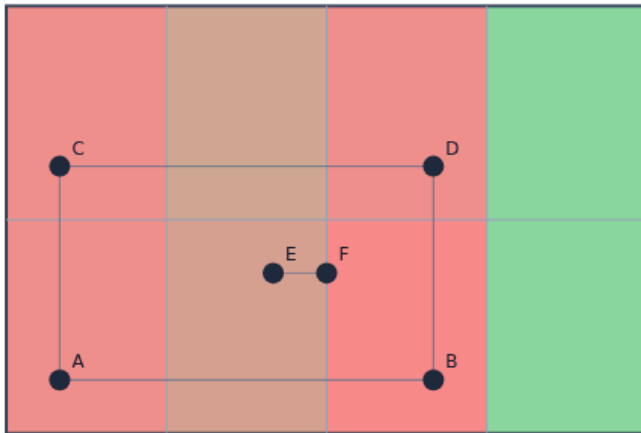
Overflow drops

# Toward routing

PLACEMENT FEEDBACK

## Toward routing

Step 5 / 5 · next · module04-01-placement-feedback



### Explanation

Real flows iterate with global routing; next course deepens GCell edges.

- learn\_global\_routing

CTS also next

Toward routing

# Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Placement feedback loop

INTERACTIVE TOOL

# Placement feedback loop

One estimate → push pass to reduce overflow on a tiny instance.

**Your job:** click *Run feedback* (or nudge) until overflow drops below the seed total (5.00).

Reset workspace

**Challenges**0 / 8 cleared

Pick one [Intro] Seed total known

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓ Reset to starter Reveal golden (study)

Run feedback

**Graph**

**Reference / metrics**

selected: A @ GCell (1,0)  
HPWL: 7.0  
overflow total: 5.00 · max: 5.00 · count: 1  
capacity: 2  
view: your placement

Browser lab starter

learn\_congestion — 01 placement feedback

# Implement track

- Implement ``placement_feedback_lite``
- Print overflow before and after on `congested_seed`
- Assert after is strictly less than before at Cap equals two

# Pitfalls

- Pushing macros that should stay fixed
- Infinite oscillation from oversized steps, use a fraction of GCell size
- Declaring victory without recomputing demand after the move

# Your turn

- Complete the feedback lab and offline compare next
- Then the wrap points you to global routing